Lab 1 — Requirements Engineering & UML Use-Case Modelling

Problem Statement #12: Healthcare & Telemedicine — Blood Bank Inventory & Emergency Donor Matcher

Deliverable 1 — Complete Requirements Table

Req ID	Type	Description	Priority	Acceptance Criteria	Rationale
FR-001	Functional	The system shall cross-reference emergency blood unit requests against real-time blood bank inventory and broadcast SMS alerts to compatible, eligible donors within a 10 km radius when sufficient stock is unavailable.	High	**Pass:** Compatible eligible donors within 10 km are notified within 30 seconds of the emergency being flagged. **Fail:** An incompatible/ineligible donor is notified or notification exceeds 30 seconds.	Enables rapid donor mobilization during critical blood shortages.
FR-002	Functional	The system shall allow the Blood Bank Manager to add, update, allocate, and remove blood units while recording blood group, component type, collection date, expiry date, and availability status.	High	**Pass:** An authorized inventory update is reflected within 2 seconds and all specified fields are stored correctly. **Fail:** Inventory displays stale or incorrect data after the update.	Accurate inventory is essential for determining whether an emergency request can be fulfilled.
FR-003	Functional	The system shall allow an Emergency Requester to submit an emergency request specifying required blood group, blood component, number of units, hospital/location, and urgency.	High	**Pass:** A request containing all mandatory valid fields receives a unique request ID and is recorded within 2 seconds. **Fail:** An incomplete request is accepted or a valid request is not recorded.	Provides structured information needed to search inventory and identify donors.
FR-004	Functional	The system shall monitor the expiry dates of stored blood components and flag units approaching expiry or already expired.	Medium	**Pass:** Units within 48 hours of recorded expiry are flagged as “Expiring Soon,” and units past expiry are automatically marked unavailable. **Fail:** An expired unit remains available for allocation.	Prevents expired blood components from being treated as available stock.
FR-005	Functional	The system shall allow notified donors to accept or decline an emergency donation request and shall provide the Blood Bank Manager with the donor response status.	High	**Pass:** A donor response is recorded and visible to the Blood Bank Manager within 5 seconds. **Fail:** The response is lost, duplicated, or associated with the wrong emergency request.	Allows the blood bank to determine whether sufficient donors have responded.
NFR-001	Nonfunctional — Performance & Security	The system shall maintain strict transactional consistency for blood inventory so that the same blood unit cannot be allocated simultaneously to multiple emergency requests.	High	**Pass:** During a concurrency test of at least 100 simultaneous allocation attempts on the same unit, exactly one allocation succeeds and all others are rejected. **Fail:** More than one request obtains the same unit.	Prevents double allocation and maintains integrity of critical inventory data.
NFR-002	Nonfunctional — Availability & Performance	The system shall provide emergency request and inventory services with at least 99.9% monthly availability and shall return inventory search results within 3 seconds under a simulated load of 100 concurrent users.	High	**Pass:** Monitoring shows ≥99.9% monthly availability and 95% of inventory searches complete within 3 seconds during the specified load test. **Fail:** Either threshold is not met.	Emergency healthcare workflows require reliable and timely access to blood availability information.

Actor summary: Emergency Requester, Blood Bank Manager, Donor, SMS Gateway.

Core workflow: Emergency Request → Check Inventory → Allocate Available Units OR Match Eligible Donors → Send Alerts → Record Donor Responses.